Riwaq: A Transparent Computational Framework for Isnād-cum-Matn Analysis and Hadith Transmission Studies

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Abstract

Hadith transmission analysis examines how oral and written historical reports circulated in early Islamic society. Modern academic hadith scholarship uses two main historical-critical approaches: common-link analysis, established by Joseph Schacht and expanded by G.H.A. Juynboll, which studies branching patterns in transmission chains (*isnāds*); and *isnād-cum-matn* analysis (ICMA), developed by Harald Motzki, which cross-checks the branching structure of transmission chains against textual variations in the reported content (*matn*). While ICMA has become one of the most effective tools for reconstructing early transmission history, conducting it manually is difficult, labor-intensive, and prone to human error when managing dozens of manuscript collections, ambiguous transmitter names, and tangled variants.

Riwaq is an evidence-bound computational research framework designed to make hadith transmission analysis accessible, transparent, and reproducible. It represents transmission families as directed graphs, calculates transparent structural metrics (such as fan-out, variant coverage, collector diversity, and bypass routes), and binds analytical claims to inspectable biographical and bibliographic citations. This paper presents the methodological foundations of ICMA, introduces Riwaq's architecture and heuristic scoring system, and demonstrates how computational tools can support both specialist

scholars and non-specialist researchers. Riwaq does not replace human scholarly judgment with an automated oracle; instead, it provides an auditable digital workbench where transmission networks and textual variants can be systematically explored, verified, and shared.

Keywords: hadith studies; *isnād*; *matn*; common link; Harald Motzki; G.H.A. Juynboll; Joseph Schacht; *isnād-cum-matn* analysis; digital humanities; evidence provenance; reproducible research; computational historiography

> **Interactive Web Application and Open-Source Repository:** Readers and researchers can explore and test Riwaq directly in their web browser without any installation at https://munir-abbasi.github.io/riwaq. The complete open-source codebase, data schemas, and reproducibility test suites are available on GitHub at https://github.com/munir-abbasi/riwaq.

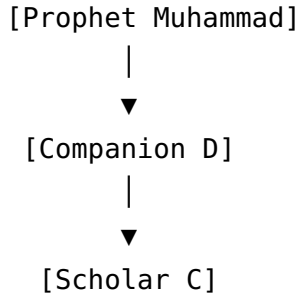
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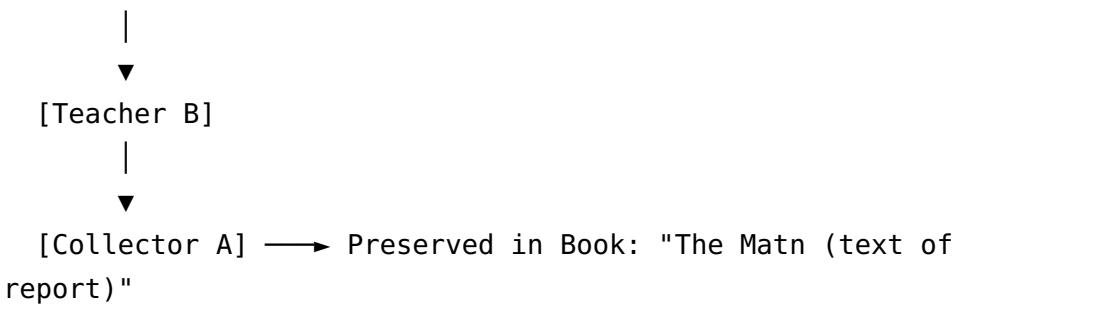
1. Introduction: The Challenge of Hadith Transmission Analysis

1.1 What is Hadith Transmission?

In Islamic tradition, a hadith is an account that preserves the statements, actions, silent approvals, or descriptions of the Prophet Muhammad, as well as those of his early companions. Each complete hadith report consists of two distinct components:

1. **The *isnād* (the chain of transmitters):** A chronological sequence of individuals through whom the report passed from one generation to the next (for example: “Collector A heard from Teacher B, who heard from Scholar C, who heard from Companion D, who heard from the Prophet”).
2. **The *matn* (the transmitted text):** The actual substantive report, speech, legal ruling, or historical narrative.





Over the first three centuries of Islamic history (seventh through ninth centuries CE), thousands of teachers taught these reports to students in major intellectual centers such as Medina, Mecca, Kufa, Basra, Damascus, and Baghdad. As teachers dictated reports to their students, and as students wrote them down or memorized them, individual reports spread across the Islamic world.

When later scholars compiled massive encyclopedic collections (such as the *Ṣaḥīḥ* collections of al-Bukhārī and Muslim, or the *Sunan* compilations of Abū Dāwūd, al-Tirmidhī, and al-Nasāʾī), they recorded multiple different chains of transmission for the same core historical event or saying. A single event might be preserved through ten, twenty, or more than fifty distinct transmission routes, each with slight variations in wording.

1.2 The Methodological Divide: Traditional Criticism and Modern Academic Studies

For over a millennium, scholars have evaluated these transmission networks, though using different questions and methodologies:

- **Classical Hadith Criticism (*ʿIlm al-Rijāl* and *ʿIlal*):** Classical Muslim scholars developed biographical evaluation (*al-jarḥ wa-l-taʿdīl*) to assess the reliability of transmitters. They scrutinized a narrator's truthfulness (*ʿadāla*), memory precision (*ḍabṭ*), teachers, students, and historical plausibility (Dickinson, 2001; Brown, 2009). They also studied hidden defects (*ʿilal*) by comparing parallel transmission routes to detect misattributions or narrative confusions.
- **Modern Common-Link Analysis (Schacht and Juynboll):** In twentieth-century Western academia, Joseph Schacht (1950) observed that when multiple chains for a single hadith are mapped together, they often converge on a single transmitter in the eighth century CE before branching out to later collectors. Schacht called this transmitter the "Common Link" (CL). G.H.A. Juynboll (1983, 2007) expanded this into an elaborate structural vocabulary of transmission bundles, spiders, and diving strands. However, both

Schacht and Juynboll often interpreted the Common Link with deep skepticism, arguing that this narrator was the original inventor or fabricator of the report.
- **Isnād-cum-Matn Analysis (Motzki):** In the 1990s and 2000s, Harald Motzki (2000, 2005, 2010) shifted the debate by showing that focusing only on the names in the chains was one-sided. Motzki argued that researchers must systematically examine the *text* (*matn*) alongside the *chain* (*isnād*). If students of a Common Link in different cities record the exact same specific wording, that wording can be traced back to their common teacher. ICMA provides a historical-critical method to test whether reports were circulating earlier than the physical books in which they are recorded today.

1.3 The Core Bottleneck: Why Manual Transmission Analysis Fails at Scale

While *isnād-cum-matn* analysis has produced compelling historical findings, it is notoriously difficult to perform by hand. Researchers face several practical hurdles:

1. **Information Overload:** Analyzing a single hadith family may require collecting dozens of variant texts across scores of Arabic source books (including canonical collections, early legal digests, biographical dictionaries, and regional histories).
2. **The "Spider Web" Problem:** Tracing dozens of overlapping transmission lines on paper produces tangled diagrams where convergence points, partial common links, and anomalous bypasses are easily misidentified or overlooked.
3. **Narrator Identity Ambiguity:** Classical texts frequently refer to narrators by short names or patronymics (such as "Sufyān", "Ḥammād", or "Ibn Jurayj"). Determining whether two occurrences refer to the exact same historical person requires careful cross-referencing of death dates, teachers, and geographic locations.
4. **Subjective Heuristics and Lack of Transparency:** Two scholars drawing trees by hand may identify different Common Links because their visual criteria and threshold rules remain implicit and unstated.
5. **The Threat of Hallucinated Citations:** In the modern era of digital databases and AI-assisted scholarship, computer-generated citations or superficial text summaries can easily introduce fake references or misattributed ratings (Ji et al., 2023).

1.4 Riwaq's Purpose and Contributions

Riwaq was created to solve these practical bottlenecks. It is a local-first, evidence-bound computational research framework that makes transmission analysis explicit, transparent, and reproducible for both specialists and everyday students of history.

This paper makes four main contributions:

1. **A Transparent Computational Heuristic for ICMA:** Riwaq translates structural concepts (such as Common Links, Partial Common Links, bundle coverage, collector diversity, and bypass routes) into inspectable mathematical features and configurable weights.
2. **An Anti-Hallucination Evidence-Binding Architecture:** Analytical claims in Riwaq must be bound to verifiable historical evidence records citing specific authors, books, editions, volumes, pages, and verbatim Arabic citations. Unbound claims are blocked from being exported as factual findings.
3. **Deterministic, Reproducible Research Workflows:** Given the same input data, Riwaq produces identical results, allowing scholars anywhere in the world to audit, replicate, or challenge an analysis.
4. **An Accessible Bridge Between Complex Data and Intuitive Understanding:** Riwaq translates dense mathematical and historical data into clear visual graphs, synoptic text comparisons, and human-readable reports in Markdown, PDF, and DOCX formats.

Riwaq is an investigative tool, not an oracle. It does not replace the critical acumen of the historian, nor does it issue final theological pronouncements. Its purpose is to provide researchers with an open, auditable digital workbench where hypotheses about hadith transmission can be tested against evidence.

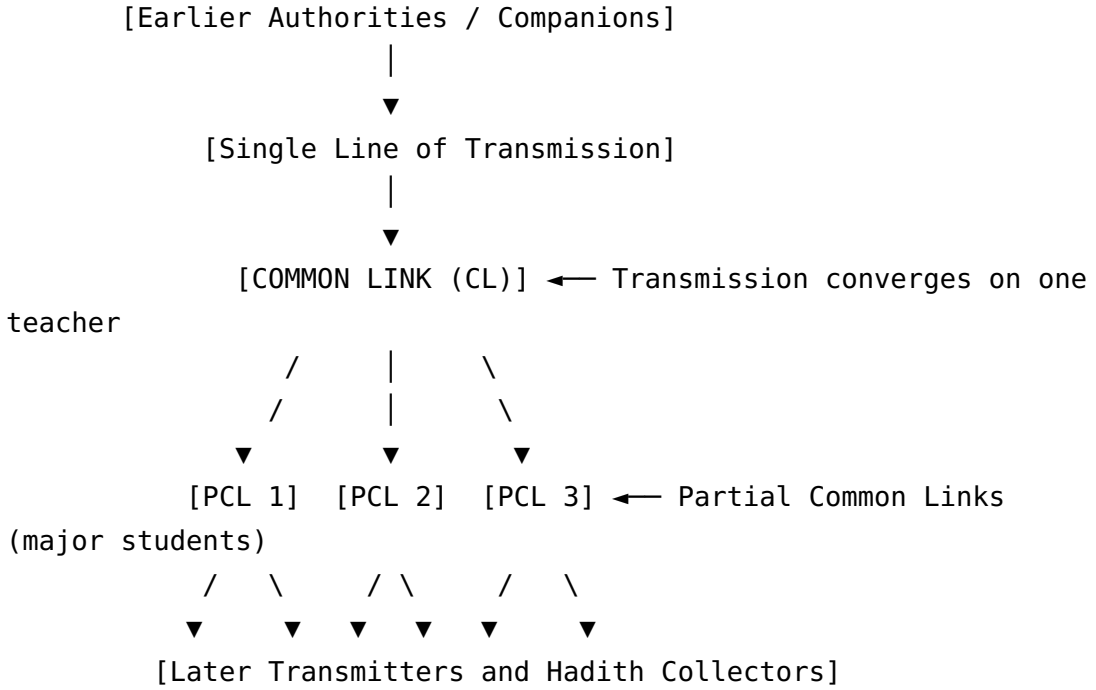
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2. Foundations of Isnād-cum-Matn Analysis (ICMA)

2.1 The Concept of the Common Link (CL) and Partial Common Link (PCL)

To understand how computational tools assist transmission analysis, one must first understand how historical reports branch over time.

Consider how a news report spread before modern telecommunications. If a teacher in Basra in the year 750 CE taught a report to five students, and each of those five students later traveled to different cities (such as Baghdad, Cairo, or Nishapur) and taught it to their own students, the resulting transmission tree resembles an hourglass:



- **Common Link (CL):** The earliest transmitter upon whom multiple independent chains of transmission converge, and from whom the report branches out to multiple independent students.
- **Partial Common Link (PCL):** A prominent student of the Common Link who passes the report to multiple students of their own, creating a secondary branching cluster.
- **Single Strand:** An unbranched transmission chain that passes from single individual to single individual without independent corroboration.

In the structural vocabulary of Schacht (1950) and Juynboll (1983, 2007), the Common Link represents the historical horizon at which a report became widely known and publicly circulated.

2.2 Moving Beyond Chain Skepticism: Harald Motzki and the "Cum-Matn" Insight

Joseph Schacht and G.H.A. Juynboll observed these branching patterns, but they drew skeptical conclusions. Juynboll argued that the Common Link was almost always the person who invented the report, and that any chains extending back from the Common Link to the Prophet were fictitious back-projections.

Harald Motzki demonstrated that this skepticism was methodologically incomplete because it ignored the actual words of the reports (*matn*). Motzki asked a simple question: **What happens when we compare the variations in the text with the branches in the chain?**

Imagine that a hadith is preserved in twenty variants:

- Ten variants go through student **PCL 1** in Kufa. All ten variants contain an extra legal clause: *"and he wiped over his leather socks."*
- Ten variants go through student **PCL 2** in Basra. All ten variants lack this clause, but all share a specific synonym for the word "fasting."

```
flowchart TD
    CL["Common Link (Teacher)"] --> PCL1["PCL 1 (Kufa)"]
    CL --> PCL2["PCL 2 (Basra)"]

    PCL1 --> S1["Student A (Text includes: 'wiped over socks')"]
    PCL1 --> S2["Student B (Text includes: 'wiped over socks')"]
    PCL1 --> S3["Student C (Text includes: 'wiped over socks')"]

    PCL2 --> S4["Student D (Text uses specific synonym)"]
    PCL2 --> S5["Student E (Text uses specific synonym)"]
    PCL2 --> S6["Student F (Text uses specific synonym)"]
```

If multiple independent students who lived in different towns and wrote their own notebooks all attribute the exact same wording to PCL 1, they could not have conspired across hundreds of miles to invent that specific wording. The most parsimonious historical explanation is that **PCL 1 really did teach that specific wording**.

Furthermore, because both PCL 1 and PCL 2 share the core body of the report, the core text must date back to their shared teacher, the **Common Link**. By correlating textual agreements and disagreements with the transmission branches, Motzki showed that historians can date traditions back to the generation of the Common Link, and often test whether the report originated even earlier.

2.3 The Classic Six Steps of ICMA

Methodologically, conducting an *isnād-cum-matn* analysis involves six consecutive steps:

1. **Exhaustive Corpus Collection:** The researcher gathers all surviving recorded instances (*turuq*) of a report from canonical collections (Bukhārī, Muslim, etc.), early musannafs (ʿAbd al-Razzāq, Ibn Abī Shayba), legal treatises, and historical chronicles.
2. **Constructing the Isnād Bundle:** The researcher reconstructs the complete transmission graph, mapping where chains converge and diverge.
3. **Identifying Common Link Candidates:** The researcher locates candidate Common Links and Partial Common Links based on their branching properties.
4. **Synoptic Text Comparison:** The researcher places the Arabic texts of all variants side-by-side, noting every addition, omission, word substitution, and grammatical variation.
5. **Correlating Textual Variations with Isnād Branches:** The researcher tests whether specific word variations cluster along specific branches of the transmission tree.
6. **Historical Synthesis and Dating:** Based on these correlations, the researcher reconstructs the earliest recoverable wording and determines the historical window during which the report circulated.

2.4 Why ICMA Needs Computational Support

Despite its analytical power, ICMA has been limited by the physical burden of manual data processing. Reconstructing a bundle of fifty variants requires tracking hundreds of transmitter relationships.

Researchers frequently encounter homonymous transmitters, uncertain transmission modalities (e.g., whether a text was heard in a formal lecture, read from a book, or transmitted by informal letter),

and anomalous chains. Without digital tools to manage this complexity, manual ICMA is time-consuming, difficult to replicate, and susceptible to confirmation bias.

Riwaq provides the systematic computational environment needed to perform these six steps with speed, mathematical clarity, and absolute evidentiary traceability.

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3. The Riwaq Computational Framework

3.1 Design Philosophy: Local-First, Deterministic, and Evidence-Bound

Riwaq is available both as an open-source research repository on GitHub (https://github.com/munir-abbasi/riwaq) and as an interactive, client-side web application hosted on GitHub Pages (https://munir-abbasi.github.io/riwaq). It is built on four core design principles:

- **Local-First Privacy:** Riwaq runs entirely within the researcher's local browser and environment. Historical texts and unpublished research notes are never transmitted to external cloud servers or third-party APIs.
- **Deterministic Execution:** The analysis is mathematically deterministic. Running Riwaq on the same input dataset with the same configuration produces the exact same scores, graph structures, and report tables every time.
- **Explicit Assumptions:** There are no opaque algorithmic black boxes. Every feature formula, normalization weight, penalty score, and classification threshold is explicitly declared in the output report.
- **Evidence-Bound Provenance:** A claim is only as good as its source. Riwaq requires that claims regarding transmitter reliability, dates, or textual readings cite specific historical evidence records.

3.2 The Core Data Model

Riwaq structures hadith transmission data into six discrete, interconnected entity types:

```
classDiagram
    class TransmissionFamily {
        +String family_id
```

```
    +String title
    +List~Variant~ variants
}
class Variant {
    +String variant_id
    +String matn
    +List~String~ chain
    +SourceWitness source_witness
}
class SourceWitness {
    +String collection
    +String volume
    +String page
    +String hadith_number
}
class Narrator {
    +String narrator_id
    +String primary_name
    +List~String~ aliases
    +Integer death_year_ah
}
class TransmissionEdge {
    +String from_narrator
    +String to_narrator
    +String modality
}
class EvidenceRecord {
    +String evidence_id
    +String scholar
    +String work
    +String citation_text
    +String source_locator
}

TransmissionFamily *-- Variant
Variant *-- SourceWitness
Variant --> TransmissionEdge
TransmissionEdge --> Narrator
Narrator ..> EvidenceRecord : backed by
```

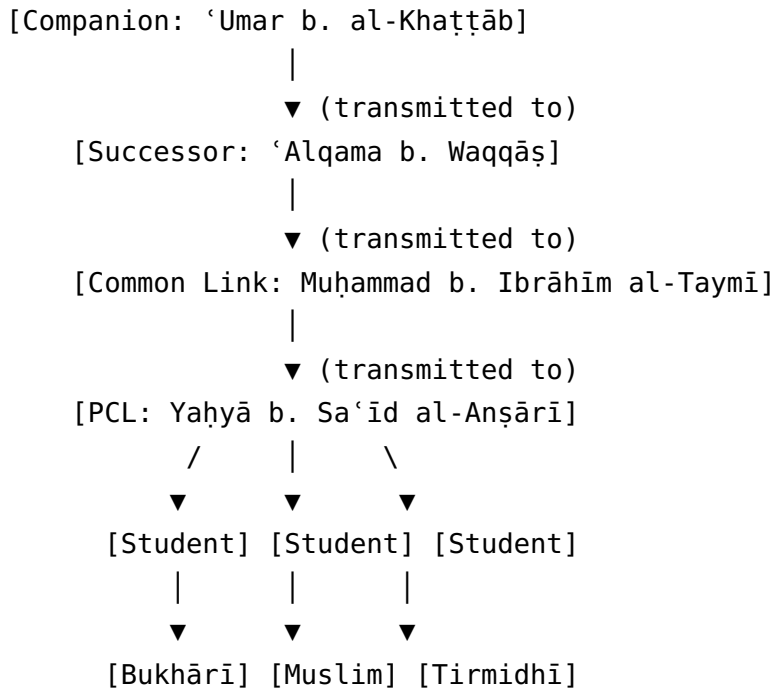
1. **Narrator:** A unique, disambiguated historical individual, carrying a stable identifier, primary name, known aliases, death year in the Islamic calendar (AH), and geographical context.

2. **Source Witness:** The physical or printed book from which a report was extracted (for example, *Ṣaḥīḥ al-Bukhārī*, hadith number 1, or *Musnad Aḥmad*, volume 2, page 150).
3. **Variant (*Ṭarīq*):** One unique pathway of a hadith recorded in a specific source witness, containing both its transmission chain and its specific textual wording (*matn*).
4. **Transmission Edge:** A directed teacher-to-student connection between two narrators, annotated with transmission modality where attested (*samāʿ*, *qirāʾa*, *ijāza*, *kitāba*, etc.).
5. **Transmission Family:** The complete bundle of related variants brought together for joint comparative analysis.
6. **Evidence Record:** An inspectable citation record linking a biographical or critical claim to a specific historical work, editor, publication year, volume, and page.

3.3 Graph Representation and Transmission Direction

Riwaq constructs a directed graph where nodes represent narrators and edges represent reported transmission relationships.

In Riwaq's graph architecture, the direction of transmission flows chronologically: **earlier authority (teacher) →\rightarrow later transmitter (student) →\rightarrow hadith collector**.



Importantly, Riwaq treats the graph as an analytical model of what the source texts *report*, rather than an uncritical mirror of historical reality. The graph allows researchers to uncover structural patterns that are invisible when reading linear text alone.

3.4 Entity Resolution: Avoiding the Identity Trap

One of the most dangerous pitfalls in manual transmission analysis is narrator confusion. In classical Arabic texts: * A single narrator may appear under multiple names, nicknames (*laqab*), patronymics (*kunya*), or geographic designations (*nisba*). For example, *Ibn Shihāb*, *al-Zuhrī*, and *Muḥammad b. Muslim al-Qurashī* are all the exact same famous scholar. * Conversely, two completely different scholars living in the same century might share the exact same name (such as *Sufyān al-Thawrī* of Kufa and *Sufyān b. ʿUyaynah* of Mecca).

If an analyst accidentally combines two distinct individuals into one node, or splits one person into two, the entire transmission tree becomes distorted.

Riwaq incorporates a dedicated entity-resolution engine (`scripts/entity-resolution.js`) that manages narrator identities through an **append-only, replayable operation log**. When two narrator entries are merged, or when an entry is split, Riwaq generates an auditable cryptographic receipt that documents: * the exact source identifiers before the change; * the rationale for the merge or split; * the preserved evidence links; and * verification that no orphan identifiers or detached evidence records remain.

This ensures that identity decisions are transparent, fully reversible, and can be inspected by other scholars.

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4. How Riwaq Analyzes Transmission Structure

4.1 De-Mystifying the Math: A Heuristic, Not an Oracle

Riwaq calculates a **Structural Support Score** for every candidate narrator in a transmission family.

It is vital to state clearly what this score is and what it is not: * **What it is:** A transparent, rule-based heuristic designed to rank and highlight narrators who exhibit the branching characteristics of Common Links or Partial Common Links. It saves researchers hours of manual counting. * **What it is not:** It is **not** a statistical probability, **not** a theological score of hadith authenticity, and **not** a divine verdict on historical truth. A score of 0.85 does not mean there is an “85% probability that the hadith is authentic.” It simply means that the narrator strongly satisfies the structural criteria defined in the algorithm.

4.2 The Five Structural Signals (S1S_1 to S5S_5)

Riwaq evaluates five structural signals for every narrator node nn. Each signal produces a normalized value between 0.000.00 and 1.001.00:

clamp(x)=max(0,min(1,x))\text{clamp}(x) = \max(0, \min(1, x))

1. Normalized Fan-Out (S1S_1, Default Weight: 0.30)

Fan-out is the raw number of distinct, direct students who received the report from narrator nn within the analyzed family:

fan_out(n)=|directStudents(n)|\text{fan_out}(n) = |\text{directStudents}(n)|

In classical isnād analysis, a narrator begins to emerge as a branching transmission hub at 3 direct students and saturates at 8 students. Riwaq normalizes this signal using an explicit anchor range [3,8][3, 8]:

S1=clamp(fan_out(n)−38−3)=clamp(fan_out(n)−35)S_1 = \text{clamp}\left(\frac{\text{fan_out}(n) - 3}{8 - 3}\right) = \text{clamp}\left(\frac{\text{fan_out}(n) - 3}{5}\right)

A candidate with 3 students receives 0.000.00; one with 8 or more distinct students receives 1.001.00.

2. Variant Bundle Coverage (S2S_2, Default Weight: 0.25)

A candidate Common Link should be present across a dominant portion of the collected variants. If a hadith is preserved in twenty variants, and nineteen of them pass through narrator nn, that narrator exhibits high bundle coverage:

S2=bundle_coverage(n)=variantsContaining(n)totalVariantsS_2 = \text{bundle_coverage}(n) = \frac{\text{variantsContaining}(n)}{\text{totalVariants}}

3. Collector Diversity (S3S_3, Default Weight: 0.15)

Branching is more significant if the chains terminate in different independent hadith collections (e.g., Mālik's *Muwaṭṭaʾ*, Aḥmad's *Musnad*, Bukhārī, and Muslim), rather than multiple chains found inside only one collector's book.

Riwaq counts the number of distinct terminal collections reachable downstream from node nn. To normalize this signal across the interval [2,8][2, 8], Riwaq uses the anchor formula:

S3=clamp(collector_diversity(n)−28−2)=clamp(collector_diversity(n)−26)S_3 = \text{clam

If downstream chains reach 8 or more distinct collections, S3=1.00S_3 = 1.00. If they reach only 2 collections, S3=0.00S_3 = 0.00.

4. Pre-Single-Strand Ratio (S4S_4, Default Weight: 0.20)

In classical Common-Link theory, a genuine Common Link typically sits at the terminus of an unbranched single line of transmission coming down from the Prophet or Companion, before the report suddenly explodes into multiple branches.

S4S_4 measures the proportion of single-strand steps upstream of candidate node nn:

S4=pre_single_strand_ratio(n)=singleStrandUpstreamHops(n)totalUpstreamHops(n)S_4 = \text{pre_single_strand_ratio}(n) = \frac{\text{singleStrandUpstreamHops}(n)}{\text{totalUpstreamHops}(n)}

If all steps above the narrator form an unbranched single strand, S4=1.00S_4 = 1.00.

5. Matn Coherence (S5S_5, Default Weight: 0.10)

When textual comparison data are available, S5S_5 measures how consistently the wording among the students of narrator nn clusters together compared to the rest of the family. In the present implementation (D017), when textual collation is absent or unmeasured, Riwaq assigns a transparent constant baseline of 0.500.50 (accompanied by the explicit diagnostic metadata

`matn_coherence_basis: 'default_constant_unmeasured'`). This avoids making unverified semantic-distance assumptions while preserving fixed, reproducible numerical weights.

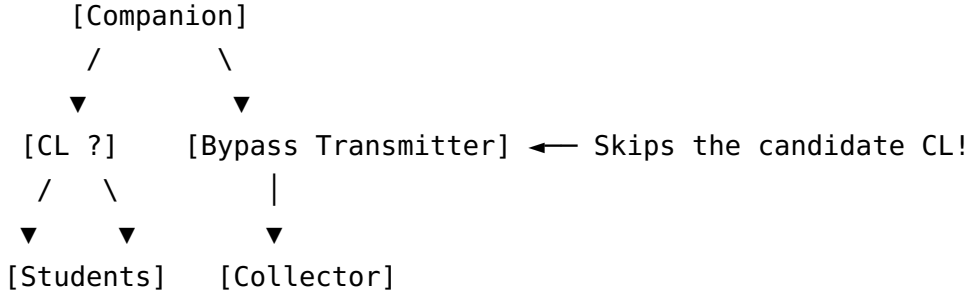
4.3 Structural Penalties (P1,P2,P3P_1, P_2, P_3)

A candidate may look structurally prominent at first glance, but possess anomalies that call their status into question. Riwaq applies three explicit deductions:

Penalty	Default Weight	Meaning	Description
P1P_1 (Bypass Ratio)	0.20	Alternative Strands	The proportion of variants in the family that reach earlier authorities while completely bypassing candidate nn.
P2P_2 (Chronology Conflict)	0.10	Historical Impossibility	Set to 1.0 if biographical records establish that teacher and student could not have met (e.g., the student was born after the teacher died); otherwise 0.0.
P3P_3 (Missing Provenance)	0.05	Incomplete Sourcing	Calculated as 1−provenance_completeness1 - \text{provenance_completeness}. Penalizes candidates whose supporting records lack proper citations.

The Importance of Bypasses (P1P_1)

A “bypass” occurs when variant chains lead to the same companion or teacher, but skip the candidate Common Link entirely. In Juynboll’s terminology, these are often related to “diving strands” or alternative routes.



A high bypass ratio indicates that the candidate was not the sole conduit for the report, meaning the candidate may only be a Partial Common Link or a regional distributor.

4.4 The Base Formula and Interpretive Bands

The **Base Score** is calculated as the fixed weighted sum of the five structural signals:

base=0.30S1+0.25S2+0.15S3+0.20S4+0.10S5\text{base} = 0.30 S_1 + 0.25 S_2 + 0.15 S_3 + 0.20 S_4 + 0.10 S_5

The final **Structural Support Score** subtracts the weighted penalties:

support_score=clamp(base−0.20P1−0.10P2−0.05P3)\text{support_score} = \text{clamp}\left(\text{base} - 0.20 P_1 - 0.10 P_2 - 0.05 P_3\right)

Candidate Detection Gates and Interpretive Bands

To prevent noise from being classified as Common Links, Riwaq enforces minimum feature gates: * **Common Link (CL) Gate:** Requires fan_out≥3\text{fan_out} \ge 3, bundle_coverage≥0.35\text{bundle_coverage} \ge 0.35, and collector_diversity≥3\text{collector_diversity} \ge 3. * **Partial Common Link (PCL) Gate:** Requires fan_out≥2\text{fan_out} \ge 2 and bundle_coverage≥0.20\text{bundle_coverage} \ge 0.20.

Candidates that pass these gates are categorized into four descriptive outcome bands:

Outcome Band	Threshold	Methodological Meaning
`supported`	≥0.75\ge 0.75	Strong structural support as a major branching node under the declared configuration.
`contested`	0.55 to <0.750.55 \text{ to } < 0.75	Meaningful structural branching, but accompanied by significant

Outcome Band	Threshold	Methodological Meaning
		bypasses or unresolved contradictions.
`uncertain`	0.35 to <0.550.35 \text{ to } < 0.55	Moderate branching, but insufficient to establish a dominant Common Link.
`likely_weak_in_context`	<0.35< 0.35	Weak structural evidence within the analyzed family; likely an ordinary link in a single strand.

The Contradiction Cap: If biographical evidence establishes a severe contradiction regarding a narrator, Riwaq enforces a hard rule: the candidate's score is capped at **0.70**, making the `supported` outcome unreachable until the contradiction is resolved.

4.5 Step-by-Step Worked Example

To see how the math works in practice, let us examine a famous historical narrator, **Hishām b. ʿUrwah** (d. 146 AH / 763 CE), analyzing a family of variants attributed to him:

1. **Calculate Signals:**
    - Hishām taught the report to 7 direct students across the family ($3 \le 7 \le 8$3≤7≤83 \le 7 \le 8): $S1 = \text{clamp}\left(\frac{7 - 3}{5}\right) = \frac{4}{5} = 0.80$S1=clamp(7−35)=45=0.80S_1 = \text{clamp}\left(\frac{7 - 3}{5}\right) = \frac{4}{5} = 0.80
    - Hishām appears in every single variant collected for this specific family: $S2 = 1.00$S2=1.00S_2 = 1.00.
    - Downstream variants reach 5 distinct hadith collections: $S3 = \text{clamp}\left(\frac{5 - 2}{6}\right) = \frac{3}{6} = 0.50$S3=clamp(5−26)=36=0.50S_3 = \text{clamp}\left(\frac{5 - 2}{6}\right) = \frac{3}{6} = 0.50
    - Above Hishām, the chain runs through his father ʿUrwah to his aunt ʿĀʾisha. This upstream path is predominantly an unbranched single line: $S4 = 0.80$S4=0.80S_4 = 0.80.

    - Textual collation is pending, so matn coherence receives the neutral constant baseline: $S5=0.50S_5 = 0.50$.

2. **Compute Base Score:** $base=0.30(0.80)+0.25(1.00)+0.15(0.50)+0.20(0.80)+0.10(0.50)\text{base} = 0.30(0.80) + 0.25(1.00) + 0.15(0.50) + 0.20(0.80) + 0.10(0.50)$
   $base=0.240+0.250+0.075+0.160+0.050=0.775\text{base} = 0.240 + 0.250 + 0.075 + 0.160 + 0.050 = 0.775$

3. **Evaluate Penalties:**

    - No bypasses exist (all variants pass through Hishām): $P1=0.00P_1 = 0.00$.
    - All teacher-student dates are historically harmonious: $P2=0.00P_2 = 0.00$.
    - All citations are fully documented with print editions: $P3=0.00P_3 = 0.00$.

4. **Final Result:**
   $support_score=clamp(0.775−0)=0.775\text{support_score} = \text{clamp}(0.775 - 0) = \mathbf{0.775}$

Because $0.775 \ge 0.750.775 \ge 0.75$, Riwaq classifies Hishām b. ʿUrwah as **`supported`**.

How to interpret this result: The algorithm indicates that Hishām b. ʿUrwah serves as a major, structurally supported Common Link in this dataset. It does **not** claim that Hishām fabricated the report; rather, it identifies Hishām as the key historical transmitter whose teaching generated the surviving textual variants.

---

5. Powering Isnād-cum-Matn Analysis in Practice

Having examined the structural scoring engine, we can now look at how Riwaq actively supports the real-world workflow of *isnād-cum-matn* analysis.

```
flowchart LR
    A["1. Ingest Variants\n(Chains + Texts + Citations)"] --> B["2. Entity Resolution\n(Disambiguate Names)"]
    B --> C["3. Graph Construction\n(Generate Isnād Bundle)"]
    C --> D["4. CL/PCL Detection\n(Calculate S1-S5 & Penalties)"]
    D --> E["5. Synoptic Text Diffing\n(Align Matn Variants)"]
```

```
    E --> F["6. Branch-Text Correlation\n(Match Wording to
Branches)"]
    F --> G["7. Evidence Binding Gate\n(Verify All Citations)"]
    G --> H["8. Export Research Dossier\n(Markdown, PDF, DOCX,
JSON)"]
```

5.1 Step 1 & 2: Assembling the Bundle and Resolving Identities

A researcher begins by inputting hadith variants into Riwaq, either via the interactive browser interface or through a structured JSON file. Each variant includes the narrator chain, the source citation, and the full Arabic *matn*.

Riwaq automatically normalizes names and flags potential ambiguities. For instance, if one variant mentions “Ibn Shihāb” and another mentions “al-Zuhrī”, Riwaq’s entity resolution links them to the canonical identifier `al-zuhri-muhammad-ibn-muslim`, preserving both original text representations while creating a single unified graph node.

5.2 Step 3 & 4: Visualizing the Bundle and Identifying Common Links

Once the variants are entered, Riwaq generates a clear interactive spider map of the transmission bundle.

Instead of squinting at tangled hand-drawn lines, the researcher can instantly see: * where the chains converge on a Common Link; * which students act as Partial Common Links (PCLs); * where single strands run; and * whether any anomalous bypass routes exist.

Riwaq runs the structural analyzer and displays candidate cards showing the fan-out, collector count, coverage percentage, and final classification band for each candidate.

5.3 Step 5 & 6: Synoptic Text Comparison and Branch Correlation

This is the heart of *isnād-cum-matn* analysis. Riwaq provides a side-by-side synoptic viewer that compares the *matn* of selected variants.

```
┌──────────────────────────────────────────────────────────────────
│ Synoptic Text Comparison
```

```
View                                                    |
├──────────────────────────────────────────────────────────────────────────
| Variant 1 (via PCL Mālik b.
Anas):                                                  |
| « إِنّمَا الأَعْمَالُ بِالـنِّيّةِ وَإِنّمَا لِامْرِئ مَا نَوَى ... »          |
|
|
| Variant 2 (via PCL Sufyān b.
ʿUyaynah):                                             |
| « إِنّمَا الأَعْمَالُ [ بِالـنِّيّاتِ ] وَإِنّمَا لِكُلِّ امْرِئ مَا نَوَى ... »   |
|
|
| Alignment
Analysis:                                                                  |
| [Shared Core]: « إِنّمَا الأَعْمَالُ ... وَإِنّمَا ... مَا نَوَى »
|
| [PCL 1 Dialect/Variant]: singular « بِالـنِّيّةِ » (Mālik's
students)      |
| [PCL 2 Dialect/Variant]: plural « بِالـنِّيّاتِ » (Sufyān's
students)     |
└──────────────────────────────────────────────────────────────────────────
```

The researcher can test specific hypotheses: * **Testing Student Transmission:** Do all five students of Mālik preserve the singular form *bi-l-niyyah*? If yes, that confirms the singular form was taught by Mālik in Medina. * **Testing the Common Link Horizon:** Do both the Medina branch (Mālik) and the Mecca branch (Sufyān) agree on the rest of the sentence? If yes, that proves the sentence was not invented by either student; it goes back to their shared teacher, Yaḥyā b. Saʿīd al-Anṣārī.

Riwaq bridges the gap between text and graph, allowing researchers to see the exact relationship between textual changes and transmission nodes.

5.4 Step 7 & 8: Exporting Auditable Research Dossiers

When the analysis is complete, Riwaq synthesizes the findings into a canonical report. The researcher can export a complete, publication-ready research dossier in Markdown, PDF, DOCX, or machine-readable JSON.

Because every chart, table, and score is generated deterministically, peer reviewers or readers can download the raw dataset, load it into Riwaq, and replicate the entire analysis in seconds.

---

6. Evidence Provenance and Grounding: Preventing AI Hallucinations

6.1 The Crisis of Plausible Fiction in Digital Scholarship

The rapid rise of large language models (LLMs) and automated research assistants has created a major crisis in academic research: **hallucination**. AI tools can generate beautifully written, grammatically flawless paragraphs that cite non-existent books, invented page numbers, and fabricated scholar quotes (Ji et al., 2023).

In hadith studies, where scholarship hinges on exact historical manuscripts and verbatim transmitter evaluations, an unverified citation destroys the integrity of an entire study.

6.2 Evidence Records: Tying Every Claim to Real Books

Riwaq prevents this problem through an architectural safeguard: **Evidence Binding** (`scripts/evidence-binding.js`).

In Riwaq, no narrator can be assigned a biographical rating (such as *thiqah* [trustworthy] or *ḍaʿīf* [weak]) unless that rating is linked to an inspectable, structured **Evidence Record**.

An Evidence Record requires verifiable bibliographical metadata:

```
{
  "evidence_id": "ev-hisham-001",
  "narrator_id": "hisham-ibn-urwah",
  "rating": "thiqah",
  "scholar": "Ibn Ḥajar al-ʿAsqalānī",
  "work": "Taqrīb al-Tahdhīb",
  "citation_text": "ثقة فقيه ربما دلس",
  "citation_span": "entry 7302, p. 573",
  "edition": "Muḥammad ʿAwwāma (ed.), Dār al-Rashīd, 1986",
  "source_type": "print",
  "source_ref": {
    "collection": "Taqrīb al-Tahdhīb",
    "source_locator": "ʿAwwāma ed. (Dār al-Rashīd, 1986), p. 573"
  },
  "ingested_at": "2026-01-15T00:00:00.000Z"
}
```

Notice that the record preserves the exact Arabic quotation (ثقة فقيه ربما دلس, meaning “Trustworthy, a jurist, occasionally practiced tadlīs”) alongside the normalized rating (`thiqah`), the editor’s name, and the exact publication details.

6.3 Validation Gates: How Riwaq Rejects Unbound Claims

Riwaq divides validation into strict hierarchical levels:

- **Blocking Errors:**
  - A claim points to a narrator ID that does not exist.
  - A claim cites an evidence ID that cannot be found.
  - A dataset contains synthetic or dummy IDs (such as `test_*` or `dummy_*`) in a production export.
  - **Consequence:** `canExport()` returns `false`. Riwaq refuses to export the analysis as an evidence-backed study.
- **Warnings:**
  - An evidence record exists, but lacks edition or page locator details.
  - Conflicting ratings are recorded across different classical critics.
  - **Consequence:** Analysis proceeds, but the claim is prominently marked as `provenance-incomplete` in all output tables.

This system guarantees that no analytical claim can pose as “evidence-backed” unless its paper trail exists and is completely open to inspection.

6.4 Preserving Historical Disagreements

Traditional hadith criticism was a lively, debated discipline, not a monolithic bureaucracy. Ibn Ḥajar, al-Dhahabī, and Ibn Abī Ḥātim frequently disagreed about the reliability of a narrator.

Many modern software tools make the mistake of averaging conflicting ratings into a single artificial number (for example, taking one “weak” and one “reliable” rating and calling the person “semi-reliable”).

Riwaq strictly forbids silent averaging. It preserves every critic’s judgment as a distinct historical witness. If critics disagree, Riwaq surfaces the disagreement directly in the uncertainty report, ensuring that the historical nuance of early Islamic scholarship is preserved.

---

7. Verification, Reproducibility, and Empirical Validation

7.1 Software Verification: What Code Verification Proves

In software engineering, verification ensures that the code executes according to its technical specification. Riwaq maintains a comprehensive test suite (exercising schema validation, entity resolution, graph generation, feature scoring, evidence binding, and export engines).

Deterministic tests confirm that: * identical datasets produce identical outputs; * edge cases (such as missing dates, circular chains, or homonyms) are handled safely without crashing; and * deployment artifacts (`academic/index.html` and `index.html`) remain byte-for-byte identical.

7.2 What Computation Cannot Prove

It is equally important to define the limits of software verification. Passing all software tests demonstrates that Riwaq's code is bug-free and mathematically consistent. **It does not prove that a historical hadith is true or fabricated.**

Computational tools cannot: 1. Prove that an ancient narrator was telling the truth. 2. Guarantee that surviving hadith collections contain every variant that was ever spoken in the eighth century. 3. Replace the historian's knowledge of Arabic idioms, cultural context, and theological debates.

Riwaq does not claim to automate historical truth. It automates the tedious, error-prone mechanics of structural analysis so that scholars can spend their time evaluating evidence rather than untangling lines.

7.3 A Roadmap for Empirical Validation

To evaluate how Riwaq performs in real scholarly settings, the Riwaq project has outlined a three-phase validation study:

- **Phase 1 (Controlled Pilot Study):** A cohort of hadith scholars analyzes matched sets of hadith families under two conditions: manual paper analysis versus Riwaq-assisted analysis. The study

measures inter-rater agreement (using Cohen's κ\kappa), time to completion, and provenance completeness.
- **Phase 2 (Longitudinal Research Study):** Scholars use Riwaq in their ongoing academic research over several months to assess software usability (via the System Usability Scale) and identify workflow bottlenecks.
- **Phase 3 (Large-Scale Corpus Testing):** Evaluating the computational stability and throughput of the system across hundreds of complex transmission families.

---

8. Discussion and Methodological Impact

8.1 Bridging Classical Islamic Scholarship and Western Academia

For decades, classical Islamic hadith scholarship and Western historical-critical scholarship operated in largely separate spheres, often talking past one another: * Western scholars often dismissed classical *isnāds* as wholesale late fabrications without examining the granular biographical evidence compiled by Muslim critics. * Traditional scholars often viewed Western common-link methods with suspicion, seeing them as speculative attempts to undermine religious tradition.

Isnād-cum-matn analysis, as championed by Motzki, created common ground by demonstrating that transmission chains and textual variants exhibit real historical correlations that cannot be explained away by cynicism.

Riwaq deepens this bridge. By accommodating classical biographical terminology (*samāʿ*, *ʿadāla*, *jarḥ wa-l-taʿdīl*) alongside modern network topology and text diffing, Riwaq provides a shared methodological vocabulary where scholars from diverse traditions can inspect the exact same data and debate conclusions transparently.

8.2 Transparency Over Authority: Why Open Algorithms Matter

In the humanities, the danger of digital tools is the "black-box effect": a computer outputs a chart or score, and users accept it uncritically simply because it was generated by software.

Riwaq rejects this model. Every calculation performed by Riwaq is open, simple, and auditable. If a scholar disagrees with Riwaq's default feature weights (for example, if they believe collector diversity should matter more than fan-out), they can adjust the weights in the configuration file, rerun the analysis, and immediately observe how the outcome changes. Riwaq democratizes transmission analysis by putting the researcher in full control of the assumptions.

8.3 Limitations and Best Practices

Researchers using Riwaq should keep three best practices in mind: 1. **Garbage In, Garbage Out:** A graph analysis is only as reliable as the corpus collected. If an analyst omits early collections or regional works, the resulting Common Link may be an artifact of missing data rather than historical reality. 2. **Text Matters:** Structural graph analysis must always be paired with careful reading of the Arabic *matn*. A high structural score is a starting point for textual investigation, not the finish line. 3. **Verify the Sources:** Evidence binding ensures that a citation exists in the dataset; it does not replace the scholar's duty to consult the cited book and ensure that the passage was transcribed accurately.

---

9. Future Horizons

The Riwaq project is actively developing new capabilities to expand digital hadith studies:

1. **Automated Arabic Text Collation:** Developing morphological and root-aware Arabic text diffing algorithms that automatically categorize variants into synonyms, grammatical shifts, additions, and omissions.
2. **Geographical and Temporal Mapping:** Visualizing transmission lines across interactive historical maps of the early Islamic world, allowing researchers to see how reports traveled geographically from Medina to Kufa, Basra, and beyond.
3. **Collaborative Evidence Repositories:** Enabling international teams of researchers to curate and cross-verify biographical evidence records in shared, version-controlled repositories.
4. **Integration with Open Scholarly Repositories:** Building direct import and export connectors for major digital Islamic libraries (such as Shamela and OpenITI).

---

10. Conclusion

Riwaq demonstrates how computational methods can bring clarity, rigor, and accessibility to one of the most intricate fields of historical inquiry. By formalizing the structural principles of *isnād-cum-matn* analysis, providing deterministic graph scoring, and enforcing strict evidence provenance, Riwaq transforms hadith transmission analysis from a daunting manual chore into an open, auditable, and reproducible discipline.

Most importantly, Riwaq demystifies this complex field for common readers and students of history. It shows that hadith transmission is not a mysterious black box, but a traceable historical network that can be visualized, measured, and verified against centuries of preserved scholarship.

References

Abu-Alabbas, B., Dann, M., & Melchert, C. (Eds.). (2020). *Modern Hadith Studies: Continuing Debates and New Approaches*. Edinburgh University Press.

Aghaei, A. (2020). The common link and its relation to hadith terminology. In B. Abu-Alabbas, M. Dann, & C. Melchert (Eds.), *Modern Hadith Studies: Continuing Debates and New Approaches* (pp. 97–128). Edinburgh University Press. https://doi.org/10.3366/edinburgh/9781474441797.003.0006

Brooke, J. (1996). SUS: A “quick and dirty” usability scale. In P. W. Jordan, B. Thomas, B. A. Weerdmeester, & I. L. McClelland (Eds.), *Usability Evaluation in Industry* (pp. 189–194). Taylor & Francis.

Brown, J. A. C. (2009). *Hadith: Muhammad’s Legacy in the Medieval and Modern World*. Oneworld Publications.

Dickinson, E. (2001). *The Development of Early Sunnite Hadith Criticism: The Taqdima of Ibn Abi Hatim al-Razi (240/854–327/938)*. Brill.

al-Dhahabī, Shams al-Dīn Muḥammad b. Aḥmad. (1963). *Mīzān al-Iʿtidāl fī Naqd al-Rijāl* (ʿAlī Muḥammad al-Bajāwī, Ed.; 4 vols.). Dār al-Maʿrifa.

Graham, S., Milligan, I., & Weingart, S. (2015). *Exploring Big Historical Data: The Historian’s Macroscope*. Imperial College Press.

Ibn Abī Ḥātim al-Rāzī, ʿAbd al-Raḥmān b. Muḥammad. (1952). *al-Jarḥ wa-l-Taʿdīl* (8 vols.). Majlis Dāʾirat al-Maʿārif al-ʿUthmāniyya.

Ibn Ḥajar al-ʿAsqalānī, Aḥmad b. ʿAlī. (1986). *Taqrīb al-Tahdhīb* (Muḥammad ʿAwwāma, Ed.). Dār al-Rashīd.

Ibn al-Ṣalāḥ, ʿUthmān b. ʿAbd al-Raḥmān al-Shahrazūrī. (1986). *Maʿrifat Anwāʿ ʿUlūm al-Ḥadīth (Muqaddimat Ibn al-Ṣalāḥ)* (Nūr al-Dīn ʿItr, Ed.). Dār al-Fikr.

Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., et al. (2023). Survey of hallucination in natural language generation. *ACM Computing Surveys, 55*(12), Article 248, 1–38. https://doi.org/10.1145/3571730

Juynboll, G. H. A. (1983). *Muslim Tradition: Studies in Chronology, Provenance and Authorship of Early Hadith*. Cambridge University Press.

Juynboll, G. H. A. (2001). (Re)appraisal of some technical terms in hadith science. *Islamic Law and Society, 8*(3), 303–349. https://doi.org/10.1163/156851901317230611

Juynboll, G. H. A. (2007). *Encyclopedia of Canonical Ḥadīth*. Brill.

al-Mizzī, Jamāl al-Dīn Yūsuf b. ʿAbd al-Raḥmān. (1980). *Tahdhīb al-Kamāl fī Asmāʾ al-Rijāl* (Bashshār ʿAwwād Maʿrūf, Ed.; 35 vols.). Muʾassasat al-Risāla.

Motzki, H. (2000). The murder of Ibn Abī l-Ḥuqayq: On the origin and reliability of some maghāzī reports. In H. Motzki (Ed.), *The Biography of Muhammad: The Issue of the Sources* (pp. 170–239). Brill.

Motzki, H. (2005). Dating Muslim traditions: A survey. *Arabica, 52*(2), 204–253. https://doi.org/10.1163/1570058053640349

Motzki, H. (2010). Whither hadith studies? In H. Motzki, N. Boekhoff-van der Voort, & S. W. Anthony (Eds.), *Analysing Muslim Traditions: Studies in Legal, Exegetical and Maghāzī Ḥadīth* (pp. 47–124). Brill.

Schacht, J. (1950). *The Origins of Muhammadan Jurisprudence*. Clarendon Press.

Stodden, V., McNutt, M., Bailey, D. H., Deelman, E., Gil, Y., Hanson, B., Heroux, M. A., Ioannidis, J. P. A., & Taufer, M. (2016). Enhancing reproducibility for computational methods. *Science, 354*(6317), 1240–1241. https://doi.org/10.1126/science.aah6168

Appendix A. Software Architecture and Release Verification Mapping

Riwaq maintains an automated test and verification suite ensuring that every release matches the specifications in this paper. The complete source code is published at https://github.com/munir-abbasi/riwaq, and the client-side application is hosted at https://munir-abbasi.github.io/riwaq:

Verification Requirement	Implementation Component	Test Enforcement
Schema Validation	`scripts/json-validator.js`	`tests/node/json-validator.test.js`
Identity Resolution & Receipts	`scripts/entity-resolution.js`	`tests/node/entity-resolution.test.js`
Structural CL/PCL Scoring	`scripts/clpcl-analyzer.js`	`tests/node/clpcl-analyzer.test.js`
Anti-Hallucination & Evidence Binding	`scripts/evidence-binding.js`	`tests/node/phase4.test.js`
Canonical Report & Export Gating	`scripts/canonical-report.js`	`tests/node/phase5.test.js`
Multi-format Exporters (MD, DOCX, PDF)	`scripts/export-*.js`	`tests/node/phase6.test.js`
Browser UI & Deploy Parity	`index.html` ↔\leftrightarrow `academic/index.html`	`npm run verify:parity`
Deterministic Pipeline Execution	`scripts/riwaq-cli.mjs`	`node scripts/riwaq-cli.mjs pipeline`

Appendix B. Minimum Evidence Record Specification

A source-backed evidence object in Riwaq must satisfy the following schema (authoritative definition in `scripts/json-validator.js`):

- `evidence_id` (String): Unique identifier (e.g., `"ev-hisham-001"`).
- `narrator_id` (String): Target narrator key (e.g., `"hisham-ibn-urwah"`).

- `scholar` (String): Classical evaluating critic (e.g., `"Ibn Ḥajar al-ʿAsqalānī"`).
- `work` (String): Classical biographical compendium (e.g., `"Taqrīb al-Tahdhīb"`).
- `edition` (String, optional): Specific printed or manuscript edition used (e.g., `"Muḥammad ʿAwwāma (ed.), Dār al-Rashīd, 1986"`).
- `citation_span` (String): Exact locator in work (e.g., `"entry 7302, p. 573"`).
- `citation_text` (String): Faithfully transcribed verbatim Arabic text of judgment.
- `rating` (String): Normalized project category (`"thiqah"`, `"saduq"`, `"daif"`, etc.).
- `normalization_note` (String, optional): Explanation of mapping rationale from source text.
- `dissent_notes` (String, optional): Recorded disagreements from other critics.
- `curated_by` (String, optional): Researcher or annotator who verified the record.
- `ingested_at` (String): ISO 8601 timestamp of record creation.

Appendix C. Canonical Interpretive Guidance

To ensure scholarly precision and avoid misinterpretation, any analytical result produced by Riwaq must be read strictly according to the following interpretive rule:

> **Positive Reading:**
> “Given the selected source corpus, the declared entity-resolution decisions, the configured heuristic parameters, and the available biographical evidence records, this candidate narrator exhibits the reported structural transmission pattern within the analyzed family.”

> **Negative Reading (Prohibited Over-Interpretation):**
> “This computational score does not establish theological authenticity, historical certainty, or conscious fabrication, nor does it guarantee that the surviving corpus contains every transmission chain that once existed.”